

Outcome-only GRPO

		<i>outcome</i>	R_i	A_i
g_1	read → write	❌	0	0
g_2	write	❌	0	0
g_3	read → diff → write	❌	0	0
g_4	rm -rf → write	❌	0	0

$$\bar{R} = 0$$

rewards identical $\Rightarrow \text{Var}_G(R) = 0 \Rightarrow A_i \equiv 0$

DEAD UPDATE — nothing to learn from the group

RLVP (process potential Φ)

		<i>outcome</i>	Φ_i	A_i
g_1	⁺¹ read → write	❌	+1	+0.5
g_2	write	❌	0	-0.5
g_3	⁺¹ read → ⁺¹ diff → write	❌	+2	+1.5
g_4	⁻¹ rm -rf → write	❌	-1	-1.5

⁺¹ credit: precondition fulfilled (read before write)

⁻¹ penalty: irreversible action (rm -rf before the task is done)

$$\bar{\Phi} = 0.5$$

Φ varies across failures $\Rightarrow \text{Var}_G(\Phi) > 0 \Rightarrow A_i \neq 0$

GRADIENT — reinforce read/diff (g_3), suppress rm -rf (g_4)